

get_status

REFERENCE

API consumer

request an on-demand device-health snapshot

Overview

`get_status` retrieves a point-in-time health and status snapshot of the reader. It is a read-only query on the device-management (MGMT) endpoint: the app publishes the command and the device publishes a correlated reply containing power, radio, clock, terminal-link, and battery information under `deviceStatus`.

Because it is read-only, `get_status` is safe to call at any time. It complements the asynchronous `heartBeatEVT` push: `heartBeatEVT` carries only a lightweight inventory-status or battery-alert summary, so use the event stream for continuous monitoring and `get_status` for an immediate, full snapshot.

Details

Property	Value
Interaction Pattern	Request / Response (app publishes a command; device publishes the correlated reply)
State Change	READ-ONLY
Required Request Fields	<code>command</code> , <code>requestId</code>
Applies To	Endpoint: Management (MGMT). Reader: RFD40 series, RFD90 series
API Version	V1.21 (API Version Reporting)
Related	get version , get current region , reboot , set operating mode

When to Use

Use `get_status` when you need an **immediate, correlated snapshot of reader health** — power source, radio activity/connection, system clock, terminal link, and battery condition — typically in a management workflow over the MGMT endpoint. It confirms device state on demand rather than waiting for the next asynchronous heartbeat.

- **Get:** the current `deviceStatus` snapshot (power, radio, clock, terminal connection, battery).
- **Use it for:** pre-flight safety checks before a state-changing command, spot health checks, and diagnostics. (`deviceStatus.radioActivity` in the response reports the activity status of the

device's radio — see Key Fields.)

- **Prefer a different command when:** you want continuous monitoring rather than a single read — subscribe to `heartBeatEVT` instead for periodic pushes, though it carries only an inventory-status or battery-alert summary, not the full power/radio/clock/terminal/battery snapshot `get_status` returns.

Topic Mapping

Message	Direction	Topic	Note
Request	App → Reader	<code><tenantId>/MGMT/clients/cmnd/<serial></code>	On the management endpoint.
Response	Reader → App	<code><tenantId>/MGMT/clients/resp/<serial></code>	Correlated reply to the request above; the in-payload <code>requestId</code> links the twin topics.

Topics follow the pattern `<tenantId>/<topic>/<serial>` — the tenant id assigned to you at provisioning (a slot you fill in, never a literal), the base topic you configure on the endpoint, and the reader's serial number, which the reader appends itself in its model-prefixed form (for example `RFD40-212735201D0053`). `App → Reader` means your app publishes and the reader subscribes; `Reader → App` means the reader publishes and your app subscribes. To watch a whole fleet rather than one reader, put the MQTT single-level wildcard `+` in the serial slot — `<tenantId>/MGMT/clients/resp/+` — as explained under [Topics and Channels](#).

Credentials are presented once at CONNECT. The `requestId` you send is echoed in the response so you can pair reply to request. The endpoint's `mqttParams` govern QoS and retain, not this command.

See [Topics and Channels](#).

Key Fields

- `requestId` (request) — your correlation token; make it unique per request. The device echoes it in the response so you can pair reply to request.
- `deviceStatus.radioActivity` (response) — the activity status of the device's radio (`INACTIVE` / `ACTIVE`).
- `deviceStatus.powerSource` (response) — how the reader is currently powered (`WALLCHARGER` / `USB` / `CRADLE` , the contract's enum); current firmware has also been reported to return `NONE` , which the enum does not allow.
- `response.code` (response) — read this to determine success vs failure (see [Response Codes](#)).

Request Payload

The request envelope is just `command` and `requestId` — there is no payload wrapper. Use the **Schema** tab for the field reference and the **Example** tab for a ready-to-send, copy-pasteable request.

Request payload

```
1 {
2   "command" : "get_status" ,
3   "requestId" : "abcd123"
4 }
```

Request payload

* required

— not applicable

Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
1	command *	string	—	get_status	Fixed command name. Used to get the reader status information.
2	requestId *	string	—	abcd123	A unique identifier for the request, allowing tracking and debugging. Echoed back unchanged in the response.

Response Payload

The response envelope is `command` , `requestId` , `apiVersion` , `response` , and an optional `deviceStatus` snapshot. On the **Schema** tab, expand `deviceStatus` (and its `terminalConnection` , `batteryStatus` sub-objects) to see every field — its type, whether it is required, its enum/units, an example value, and a description; the **Example** tab shows the full JSON. `deviceStatus` is optional at the envelope level; which fields appear depends on device state and model — the **Example** tab shows a representative snapshot.

Response payload

```
1  {
2    "command": "get_status",
3    "requestId": "abcd123",
4    "apiVersion": "V1.21",
5    "deviceStatus": {
6      "powerSource": "USB",
7      "radioActivity": "INACTIVE",
8      "radioConnection": "DISCONNECTED",
9      "systemTime": "2026-06-15T09:26:00.443Z",
10     "ntp": {
11       "offset": 0,
12       "reach": 0
13     },
14     "terminalConnection": {
15       "status": "CONNECTED",
16       "type": "USB"
17     },
18     "batteryStatus": {
19       "mfgDate": "22DEC2024",
20       "cycleCount": 8,
21       "fullChargeCapacity": 5739,
22       "temperature": 24,
23       "designCapacity": 6400,
24       "batteryType": "LI-ION",
25       "capacity": 2467,
26       "stateOfHealth": "GOOD",
27       "chargePercentage": 46,
28       "chargeStatus": 1
29     }
30   },
31   "response": {
32     "code": 0,
33     "description": "Success"
34   }
35 }
```

Response payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
1	command *	string	—	get_status	Echoes the executed command
2	requestId *	string	—	abc123	Echoes the request's request ID — pairs reply to request.

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
3	apiVersion *	enum	V1.0 V1.1 V1.2 V1.21 V1.22	V1.21	API version the device reports
4	deviceStatus	object	—	—	Status snapshot; optional at envelope level. Which deviceStatus fields appear depends on device state and model — see the Example.
5	powerSource *	enum	WALLCHARGER USB CRADLE	CRADLE	The source of power for the device. The contract's enum WALLCHARGER, USB or CRADLE. Current firmware has also been reported to return NONE (no external power source attached which is outside the enum).
6	radioActivity *	enum	INACTIVE ACTIVE	ACTIVE	The activity status of the device radio.
7	radioConnection *	enum	CONNECTED DISCONNECTED	CONNECTED	Connection status of the RF radio.
8	hostname	string	—	RFD40-212735201D0053	Device hostname on the network. Optional; may be absent.
9	systemTime *	string	date-time	2022-07-01T03:01:31.464Z	Device system time in ISO 8601

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
10	temperature	integer	—	32	Device temperature (distinct from the battery temperature below).
11	ntp	object	—	—	Details about the Network Time Protocol (NTP) synchronization.
12	offset	integer	—	120	The offset in milliseconds between the device's clock and the NTP server. Declared in schema, but current firmware does not populate it — the value shown is the schema's own example, not a device reading. See Response Semantics.
13	reach	integer	—	321	The reachability score of the NTP server. Declared in the schema, but current firmware does not populate it — the value shown is the schema's own example, not a device reading. See Response Semantics.
14	terminalConnection	object	—	—	Host-terminal link to the selected terminal.
15	status	enum	CONNECTED DISCONNECTED	CONNECTED	Terminal connection status.
16	type	enum	BLUETOOTH CIB USB	USB	Terminal transport type.

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
17	bat ter ySt atu s	object	—	—	Battery status information.
18	m f g D a t e	string	—	22DEC2024	Battery manufacture date.
19	c y c l e C o u n t	integer	—	120	Completed battery charge cycles.
20	f u l l C h a r g e C a p a c i t y	integer	—	3200	Current full-charge capacity.

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
21	temperature	integer	—	32	Battery temperature (distinct from the device temperature above).
22	design capacity	integer	—	7000	Designed battery capacity.
23	battery type	string	—	LI-ION	Battery chemistry / type.

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
24	capacity	integer	—	5000	Remaining battery capacity.
25	stateOfHealth	enum	GOOD CRITICAL MODERATE	GOOD	Long-term battery capacity / health. The same value set the <code>batteryAlert.stateOfHealth</code> carried by alerts and <code>heartBeatEVT</code> — one shared vocabulary, so nothing need converting. See Response Semantics.
26	chargePercentage	integer	—	23	Percentage of charge remain

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
27	chargingStatus	integer	—	1	Charging status: 0 = charge connected; 1 = charger connected, charging; 2 = charger connected, battery at 100%.
28	response *	object	—	—	Result wrapper.
29	code *	integer	0–101	0	0 = Success; non-zero = error. Required by the contract. See Response Codes below.
30	description *	string	—	Success	Human-readable status.

Response Semantics

- Which `deviceStatus` fields appear depends on device state and model — absent fields are normal behavior, not an error. Note that `deviceStatus` reports two distinct temperatures: `deviceStatus.temperature` (device) and `deviceStatus.batteryStatus.temperature` (battery) — do not conflate them.
- The reader has no RTC backup battery. After a cold start or factory reset, `systemTime` starts at a baseline default; the device syncs its clock from **network time** once it has a network connection. There is currently no API to change the NTP configuration — set the NTP server details (and, if NTP is unreachable, the device time itself) from the **123RFID Desktop** application. Time-sensitive operations (TLS certificate validity, log/event timestamps) may use the default time until the clock has synced.
- `stateOfHealth` is one grade, on two surfaces. `deviceStatus.batteryStatus.stateOfHealth` here and the `batteryAlert.stateOfHealth` carried by `alerts` and `heartBeatEVT` use the **same** enum — `GOOD` / `CRITICAL` / `MODERATE` — so shared battery logic needs no conversion. Keep it distinct

from the event's `status`, which is a charge band (`LOW` / `CRITICAL` / `CHARGING` / `FULL` / `HIGH`), not a health grade. See Interpret `stateOfHealth`.

- `ntp.offset` and `ntp.reach` **are not reported by the reader**. The schema declares both fields; the values on the **Schema** tab (`120` and `321`) are the schema's own field examples, and the contract's message example shows `0` for each. Current firmware has been reported not to populate either one, so treat their absence as normal and do not build clock-health logic on them. Removing them from the schema is an upstream request to the API owners; until then the fields remain declared.

Response Codes

`get_status` returns `response.code`. The codes applicable to this command are below; for the full table see the [Response Codes](#) and [Correlation and Request/Response Pairing](#).

Code	Description
0	Success — status retrieved and returned.
3	Not able to retrieve information — the reader could not gather the requested status at this moment. Retry; if persistent, reboot after stopping any active inventory.
101	Default error — generic catch-all.

Examples

`get_status` is a single request/response exchange. Publish the standard `{command, requestId}` request shown under [Request Payload](#); the device replies on the response topic with the correlated reply — the `requestId` you send (`abcd123`) is echoed back so you can pair reply to request. The full reply, field by field, is shown under [Response Payload](#) above.

Notes and Caveats

- `get_status` is the on-demand poll; `heartBeatEVT` and related alerts push a lighter periodic summary (an inventory-status or battery-alert block), not the full `get_status` snapshot. Prefer the events for continuous monitoring and `get_status` for a complete point-in-time check. Cross-command `apiVersion` reporting can differ (for example `mqttConnEVT` emits a numeric `apiVersion`); that is a firmware note, not a `get_status` schema difference.

Related:  [The Configuration Model](#) ·  [Response Codes](#) ·  [MQTT API](#)